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RESEARCH ARTICLE

Reassessing the Education–FDI–Trade Nexus for a Sustainable Future: Evidence from Sub-Saharan Africa

[version 1; peer review: 1 approved with reservations]

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Abstract

Background

Education, foreign direct investment (FDI), and trade openness can influence environmental sustainability through competing human-capital, technology-transfer, scale, composition, and pollution-haven channels. Evidence for Sub-Saharan Africa remains fragmented and often overlooks cross-sectional dependence and asymmetric responses.

Methods

This study examines a panel of Sub-Saharan African economies from 1990 to 2019 using production-based and consumption-based carbon dioxide emissions as environmental outcomes. The empirical strategy applies cross-sectional dependence and slope-heterogeneity diagnostics, second-generation unit-root and cointegration tests, dynamic seemingly unrelated regression, continuously updated fully modified and bias-corrected estimators, asymmetric decompositions, and panel causality tests.

Results

Education is consistently associated with lower emissions across the principal long-run specifications. FDI is generally associated with lower production-based and consumption-based emissions in the symmetric long-run estimates, while the asymmetric findings show that the environmental response can differ between positive and negative FDI shocks. Trade openness is predominantly emission-increasing, and urbanisation is associated with higher emissions. The

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robustness analyses support the stability of the core associations.

Conclusions

Sustainability policy in Sub-Saharan Africa should combine environmental education with screening and performance standards for FDI and trade-related production. Policies should account for asymmetric adjustment and economy-specific conditions when pursuing low-carbon development.

Keywords
education; foreign direct investment; trade openness; environmental sustainability; carbon emissions; Sub-Saharan Africa; panel data; asymmetry

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Introduction

The ecological status of the planet has never been more susceptible to sustainable managing practices as it is at this moment in the history of modern society, facing an emergency of unparalleled scales. It is a common occurrence on all levels, and education can be a good channel to be used to mitigate it (Kumar et al., 2023). The instability and value of the financial rapport between educational pursuits and the sustainability of the environment explains the need to establish a more sustainable future of the world. The crisis of CO2 emissions to the atmosphere is one of the most significant problems of the twenty-first century that affects the economic growth, health of people, and environmental sustainability on a global scale. The major factors of carbon dioxide emissions are the anthropogenic ones especially deforestation, industrial activities and burning fossil fuels (Lone & Bhat). It is clear that emissions of the burning of fossil fuels are one of the number one sources of greenhouse gases (GHGs) and that the emission of CO2 into the atmosphere is the contributor to global warming, storms that have increased in number, sea level rising and disrupted ecological stability, which are substantial contributors to climate change (Soeder and Soeder, 2021). These changes are not limited to the environmental impact, but the same influences the social and economic fields. Furthermore, since the climate system is interconnected, the carbon dioxide emissions in a specific area will inevitably have an effect in other regions; the CO2 in the atmosphere affects all countries, no matter what the political boundaries are. In turn, it is crucial to understand that the effects of domestic emissions go global, and such phenomenon as carbon leakage also needs to be taken into consideration (Pan et al., 2020). To start solving the environmental concern of excess emission of carbon dioxide, nations have to examine the relationships between the carbon emissions, the foreign direct investment, trade openness, and education.

The complex factors underlying carbon dioxide emissions are closely intertwined with economic, ecological, and social processes. The constituent components that contribute to the overall carbon footprint do not make equal contributions. Among them, education (EDU) (Afolayan et al., 2020) and foreign direct investment (FDI) (Huang et al., 2022) are the most relevant. The other impact of emissions can be influenced by capital flows, such as FDI, which is closely related to increased economic activity. Through educational efforts, much is done to advance society's trustworthiness and progress towards greater sustainability. There are also preconditions for the intensification of emissions due to industrial and commercial growth, trade openness (TO) (Jun et al., 2020), and economic growth (EG) (Wang et al., 2022). Their energy mix significantly influences countries' emission profiles; countries with resources tend to use fossil fuels. Urbanisation (UR) can boost economic growth, but it can also increase energy use (Ali et al., 2020). Ecological footprint (EF) (Ali et al., 2020) and technological development (TD) (Xu and Lan, 2023) are also represented as emission patterns. The regulations (RQ) and institutional structures (IP) of a specific country will determine the ability to develop and adopt policies that enhance environmental conditions (Huang et al., 2022; Ali et al., 2020). Although the effect of outward foreign direct investment (OFDI) on a host country is disputed (some researchers may find potential advantages in the form of economic growth and stabilisation in the case of influence on main spheres, Zafar et al., 2022), its influence on emissions remains. The current analysis, therefore, asks whether it (EDU) is the mechanism for environmental sustainability, and whether FDI and trade openness (TO) are. The areas of investing in education, FDI, and trade liberalisation are interconnected and aimed at overcoming the difficulties faced by scholars and policymakers (Aust et al., 2020). Recent literature shows that investment in education enhances environmental safety, and FDI, as well as openness to trade, can contribute to environmental degradation (Tiba and Belaid, 2020). Providing more evidence, education (EDU) has become one of the determinants, as it has an influential effect on society and the economy.

Education (EDU) has become a determining variable with a substantial impact on the economy and society. The studies by Afolayan et al. (2020) and Xu and Lan (2023) were aimed at investigating the relationship between education and CO2 emissions. Education enhances human capital, allowing people to make informed, environmentally friendly choices. Afolayan et al. (2020) find that there is a positive correlation between elevated education and economic activity, which may also raise emissions, and the education-emission nexus is a multidimensional process. According to Zafar et al. (2022), education is important to boost the integration and use of modern technologies to achieve energy efficiency and minimize carbon emissions. The nature of jobs and employment opportunities is also influenced by education-related events (Suman, 2021). Another major cause of CO2 emissions in resource-abundant countries is foreign direct investment (FDI), as Y. Huang (2022a) and Q. Yang, Li, and Wang (2023) explain in the literature. At the same time, FDI may also lead to increased emissions, both due to economic development and the implementation of energy-efficient technologies and practices (To et al., 2019; Hanh et al., 2020). It can have a significant effect on the FDI multiplier in terms of emissions, as trated by Hao et al. (2021), and will require, base consideration oforces, type oinvestment types The research variables in Q. Wang and Zhang (2021) and Jun et al. (2020) are trade openness (TO), which can stimulate industrialisation and economic activity, thereby facilitating a rise in emissions.

However, free trade will facilitate the transition to sustainable finance and show the masses environmentally friendly technologies as well. Wang and Zhang (2021) recognize that trade policies can mitigate the effects of emissions because they influence traded goods. The looming environmental issues in Sub-Saharan Africa (SSA) have heightened the

urgency of the research. Economic growth and environmental protection must be in a delicate balance because the region is focused on economic growth. The main predictors of this balance are education, foreign direct investment (FDI), and trade openness. The interaction between these factors is the focus of the analysis, which is presented in detail to enhance the quality of policies. It is anticipated that the study will fill gaps in the literature by critically analyzing the relations using advanced econometric tools and thus provide subtle and precise information.

Although empirical studies on environmental sustainability are increasingly numerous, research findings on the roles of education, foreign direct investment, and openness to trade remain dispersed and fragmented. The literature documents the emission-cutting and emission-enhancing effects of these organizations, and this lack of certainty hinders policy-makers from developing consistent policies that can balance development and environmental conservation, especially in developing nations. Educationists and environmentalists who have researched the convergence between education and environmental sustainability have arrived at varied conclusions. Some researchers say that more education results in increased consumption and activity among people, thereby contributing to greater carbon emissions. According to other studies, education enhances environmental awareness, the adoption of cleaner technologies, and the overall level of public support for environmental protection, leading to a decrease in pollution. These are mutually conflicting arguments that are rarely addressed in a single empirical model, and most studies assess education on its own without considering other macroeconomic variables. The same discrepancies prevail in the foreign direct investment literature. The pollution haven hypothesis assumes that FDI will lead to increased emissions because more polluting industries will be transferred to countries with weaker environmental regulations. The pollution halo hypothesis, on the other hand, tends to focus on the movement of technology, efficiency gains, and the spread of superior management practices, leading to less pollution. Each of the two positions is supported by empirical data, often based on sample selection, estimation techniques, or control variables. However, it is not the subject of most research to determine the interactions among the environmental impact of FDI, human capital, and trade exposure.

The results of trade openness also provide conflicting empirical results. The scale effect presupposes greater emissions because increased production and transportation are expected. The effects of composition and technique imply that openness may lower emissions by shifting production towards cleaner industries and spreading advanced technologies. This is because in most empirical literature, the mechanisms are tested individually and based on average effects, which conceal the possibility of asymmetries and context-dependent effects.

The main literature gap is the lack of an integrated analytical approach that simultaneously examines education, FDI, and trade openness and addresses issues of heterogeneity, asymmetry, and cross-sectional dependence. These mostly used single-equation models or first-generation panel estimators, resulting in studies that captured complex, interdependent effects within countries and identified both positive and negative effects. Subsequently, conflicting results persist, and policy interpretations remain poor.

The gap this research aims to fill is the lack of a single empirical approach that combines education-based human capital, foreign direct investment, and trade openness for environmental sustainability. The framework explicitly leaves room for asymmetric responses and second-generation panel methods for long-run dynamics. In this way, the research balances theoretical debates and provides additional evidence on the circumstances in which education, FDI, and trade can contribute to environmental enrichment or degradation. The strategy is especially applicable to Sub-Saharan Africa, where rapid economic integration and human capital development are occurring alongside rising environmental pressures.

Literature review and hypothesis development

Theoretical foundation

Environmental sustainability is a discourse that is becoming increasingly vital and will continue gaining momentum today and in the future. It expresses the urgency of being efficient in marshalling resources and, at the same time, protecting the natural ecosystems on which humanity depends. In its conventional sense, environmental sustainability (ES) refers to the effort to ensure that the current generation can fulfill its needs today and that future generations will have the capacity to fulfill theirs. This knowledge is grounded in the broader concept of sustainable development, which, in turn, requires incorporating environmental stewardship, social equity, and economic development. Natural resources and ecosystems play a central role in society's welfare, and their assessment is directly linked to the community's overall well-being (Ziaul & Wang, 2023).

Empirical studies make conceptually consistent explanations of the determinants of ES, and are deeply rooted in economic theory. The Human Capital Theory teaches us that the more knowledge a person is provided with, the better prepared he/she is to handle the environment. A highly educated population is thus better acquainted with sustainable

practices and is willing to adopt new environmental protection strategies in a positive manner (Mourad, Wahid, Alkubise, and Najar, 2023). Neoclassical Environmental Kuznets Curve (EKC) theory argues that economic growth has not inevitably led to environmental degradation, as long as enlightened citizens steer countries toward cleaner technologies and a more sustainable path of development. Studies conducted by Duodu et al. show how education can influence attitudes and behaviour, with a view to facilitating civic participation in environmental campaigns, as explained in terms of behavioural economics (Ekins, 2011).

Environmental sustainability is affected both positively and negatively by the impact of trade openness. The Scale Effect means that, as industrial production and trade increase, pollution and resource consumption rise (Varghese, 2024). On the other hand, a change in the composition of the environment can be observed as the shift to cleaner industries or services producing what has been called the Environmental Composition Effect. Rehman, Bhatti, Kraus, and Ferreira (2020) contend that stringent environmental standards might complement trade, and as a result, a race to the top may mobilize in which high-income, environmentally neutral goods are swapped. There is an elegant correlation between ES and foreign direct investment (FDI). Duodu, Kwarteng, Oteng-Abayie, and Frimpong (2021) propose the Pollution Haven Theory, which suggests that multinational companies will set up in jurisdictions with weak environmental policies, thereby increasing pollution in the area. Conversely, the Pollution Halo Hypothesis states that FDI might also be beneficial to developing states by introducing new technology and improving environmental management systems (Gyimah, Fiati, Nwigwe, Vanessa, and Yao, 2023). Governance, in this respect, plays a critical role, since host-country regulations are the main drivers of the net effect of FDI on sustainability.

Urbanisation does give opportunities and challenges. The rapid urban development increases per capita pollution and resource exploitation, particularly in poorly planned cities (Bordun, Antonyak, Dumych, Hrytsyshyn, and Bilous, 2023). However, cities have economies of scale that help enhance social services and minimise waste in infrastructure (Bowles, Boetto, Jones, and McKinnon, 2016). City centres of higher learning and innovation encourage sustainability activities and nurture favourable laws and culture. Nonetheless, these objectives should be achieved through sustainable urban planning and policy frameworks (Ang, Choolani, and Poh, 2024). The four factors connected to education, trade, openness, and urbanisation interact in forming complicated systems to impact environmental sustainability. These dynamics are important to study to identify viable ways to support the provision of ecosystem services. The future governance should adopt technologies that can realise these strategies.

- Education and environmental sustainability

Education is a catalyst of growth and prosperity as it is an investment into individuals and the society in general. It has been known since long as an important factor of human development and economic growth. Over the recent years, the two-way communication between the environment and education has received the growing interest of scholars. Further elaboration of this relationship shows that it is a prerequisite to sustainability in practice, technology innovation, policy advocacy, diversify economy and exchange of best practice throughout the globe. The procedure highlights the immense ability of education to influence cognition as well as create a globally conscious society that is sustainably aware of everyone. However, there are two empirical studies, which present opposing results: the first one shows that education has a positive relationship with emissions and the other one indicates the opposite.

The quantitative data of the first group illustrate that education and CO 2 emissions have a positive correlation. Afolayan et al. (2020) have made the conclusion that a high level of education allows the economy to be active, which in turn pushes the size of the aggregate emissions. In the same fashion, both education and economic growth were found to have a positive impact on CO 2 emissions, as reported by Xu and Lan (2023). Continuous air pollution may be explained by the further use of fossil fuels. Suman (2021) suggested that the prevalence rates of greenhouses depend on the level of education and job opportunities on the regional level. Moreover, Zafar et al. (2022) hypothesized that education is also one of the impacts major causes of environmental degradation. The latter group highlights the adverse relationship between education and CO 2 emission. Leal Filho et al. (2019) noted that energy sustainability is a top priority agenda among universities in an effort to limit carbon emissions and lower energy use in buildings. These sustainability programs form a huge investment by many institutions of higher learning. As illustrated by Yin, Jiang, Lin, and Liu (2022), online education has a twofold effect on carbon emissions: to begin with, online classes reduce commuting-related emissions since most of them can be taken at home; to conclude, online education leads to emission through the use of devices and servers. Chen et al. (2022) argued that education enabled personal adaptation of the climate change and the creation of technologies that reduce the emission of greenhouse gases. According to Zafar et al. (2020), education has a positive effect on carbon emission reductions.

- Foreign direct investment and environmental sustainability

Two different dichotomies have been developed in the literature with respect to the nexus between foreign direct investment (FDI) and the greenhouse gas emissions. An extensive body of literature records a favorable relationship, that is, increased FDI inflows are correlated with emissions of carbon dioxide. According to Yuan Wang and Huang (2022a), a positive initial shock in short-term increases in augmentations in FDI can trigger an increase in CO 2. Expanding on the pollution haven hypothesis, Huang et al. (2022) argue that FDI is more concentrated in the jurisdictions where environmental regulations are relatively lenient, thus adding up to 525 percentage points of emissions by 2100, depending on geographic and institutional factors. According to Hao et al. (2021), this effect is explained by the emissions produced during the different levels of production and consumption. It is important to mention that the impact of the environment is considerably softened by the investor sphere, techno, and motivation. Also, Huang et al. (2022) emphasise the role of macro-variables namely economic development, financial development and urbanisation, and tourism in mediating the indirect impact of FDI on carbon emissions. The (quantitative) relationship was measured by Wang et al. (2022), and the coefficient estimates the result of a 1 percent increment of FDI resulting in a 0.03 percent increment in emissions; however, the error-correction coefficient has a negative value, which means that, in the short term, the system aims at returning the equilibrium situation to its original condition.

On the other hand, a substantial body of research suggests that FDI may lead to technological spillovers that reduce carbon production. To et al. (2019) showed that FDI contributes to the spread of clean technology innovations, thereby helping prevent emissions. They argued that host countries with higher education disproportionately benefit from the environmental returns of such inflows. Q. The same argument was supported by Wang et al. (2023), who reported that in better-quality countries, there is a higher likelihood that there would be a reduction in the level of CO 2 emission that can be linked to FDI. Hanh et al. (2020) also found that FDI tends to reduce pollution, primarily by providing environmentally friendly technologies that decrease energy consumption and the intensity of carbon emissions. As stated by Apergis, Pinar, and Unlu (2023) and Ren, Yuan, Ma, and Chen (2014), the potential of FDI to promote the proliferation of clean technology could be decisive in mitigating greenhouse gas emissions, and policy frameworks aimed at capitalising on positive externalities are thus critical.

- Trade openness and environmental sustainability

Two research teams reported contrasting results: one team obtained a positive result, while the other obtained a negative result. The evidence from the former indicates a positive correlation between trade openness and carbon dioxide emissions. Jun et al. (2020) postulate that high levels of trade openness may strengthen the dominance of specific industries, thereby leading to high pollution levels. Chhabra, Giri, and Kumar (2023) reveal that the production of goods can serve as an indicator of carbon emissions. Additionally, the exchange of products and services can significantly impact the adoption of environmentally friendly technology, helping countries transition to more sustainable economic activities. Nonetheless, when goods are produced in consideration of contaminated products, environmental degradation occurs. There is even a possibility that trade openness can reduce carbon emissions in the wealthier nations under the conditions described by Wang and Zhang (2021).

Furthermore, Ali et al. (2020) found that liberal trade policies can predispose an environment to degradation because nations may loosen their regulations in pursuit of trade and investment. By liberalisation, energy-intensive industries will be forced to relocate to places with less stringent environmental regulations, thereby increasing carbon emissions into the atmosphere. Two distinct claims support this relocation trend, suggesting a common underlying causal mechanism. Another academic research stream introduces a negative dependent correlation between trade openness and carbon dioxide levels. Several reports argue that liberalised trade may reduce carbon emissions in developed countries. For example, Karedla, Mishra, and Patel (2021) found that CO2 emissions in India decreased significantly as the country opened up to trade. Accordingly, Lin and Umetsu note that carbon dioxide emissions are expected to decline in newly industrialised countries as they become more open. According to Chhabra, Giri, and Kumar (2022), increased trade facilitates the exchange of knowledge and technology, thereby improving environmental performance by reducing carbon dioxide emissions. Lastly, Thi, Tran, and Nguyen (2023) argue that emissions can be reduced by utilising renewable energy sources, increasing trade openness, and fostering innovation. A summary of the literature is presented in Table 1.

Figure 1 presents the theoretical foundation of the study, and Table 1 provides the summary of the literature survey.

Table 1. Summary of literature survey.

Author	Sample (year)	Methodology	IV(s)	EDU	FDI	TO
Q. Wang, Yang, Li, and Wang (2023)	67 countries (1990–2019)	FMOLS			-VE	
Thi, Tran, and Nguyen (2023)	53 nations (19902–2019)	FMOLS, DOLS, GMM System estimator	REC, TO, INNO			-VE
Chhabra, Giri, and Kumar (2023)	BRICS (1991–2019)	DCCE	IQ, PS, PE			+VE
Apergis, Pinar, and Unlu (2023)	BRICS (1993–2012)	GMM	-		-VE	
Xu and Lan (2023)	China (2010–2021	DEA	TD	+VE		
Y. Wang and Huang (2022)	East Asia (2011–2020)	ARDL	TR, EG		+VE	
Zafar, Saleem, Destek, and Caglar (2022)	22 top remittance-receiving countries (1986–2017)	second-generation unit root techniques	remittances, export diversification	+VE		
Chen et al. (2022)	Global countries	Econometric models	-	-VE		
Chhabra, Giri, and Kumar (2022)	23 middle-income countries (1994–2018)	GMM	Inn			-VE
Hanh et al. (2020)	Vietnam and Indonesia (1960–2018)	ARDL	-		+VE	
Yin, Jiang, Lin, and Liu (2022)	China (2011–2019)	Regression model	-	-VE		
Huang et al. (2022)	G20 economies (1996–2018)	FGLS	ED, FD, UR, and RQ	+VE	+VE	
Hao, Ba, Ren, and Wu (2021)	China (2003–2016)	SYS-GMM	TR, OFDI		+VE	
Q. Wang and Zhang (2021)	182 countries (1990–2015)	Econometric models	-			-VE
Lin and Umetsu	NIC countries (1971–2020)	Econometric models	-			-VE
L. Wang et al. (2022)	NICs, (1990–2016)	ARDL-PMG	-		+VE	
Suman (2021)	Nepal	Econometric models	-	+VE		
Zafar, Shahbaz, Sinha, Sengupta, and Qin (2020)	OECD (1990–2015)	Second generation methodologies	NRA, FDI, EG	-VE		
Jun, Mahmood, and Zakaria (2020)	China	wavelet-coherence analysis	-			+VE
Ali, Yusop, Kaliappan, and Chin (2020)	OIC countries	DCCE, MG, PMG	TO, UR, EF, IP			+VE
Afolayan, Okodua, Oaikhenan, and Matthew (2020)	Nigeria (1980–2017)	ARDL	ED	+VE		
To, Ha, Nguyen, and Vo (2019)	Asian region (1980–2016)	DOLS, FMOLS	-		-VE	
Ren, Yuan, Ma, and Chen (2014)	China (2001–2011)	environmental regulation	EI, TO, CA, ER,		-VE	

Note: EDU = education; FDI = foreign direct investment; TO = trade openness. Signs summarise the direction of the reported relationship with environmental sustainability. Source: authors' compilation from the cited literature.

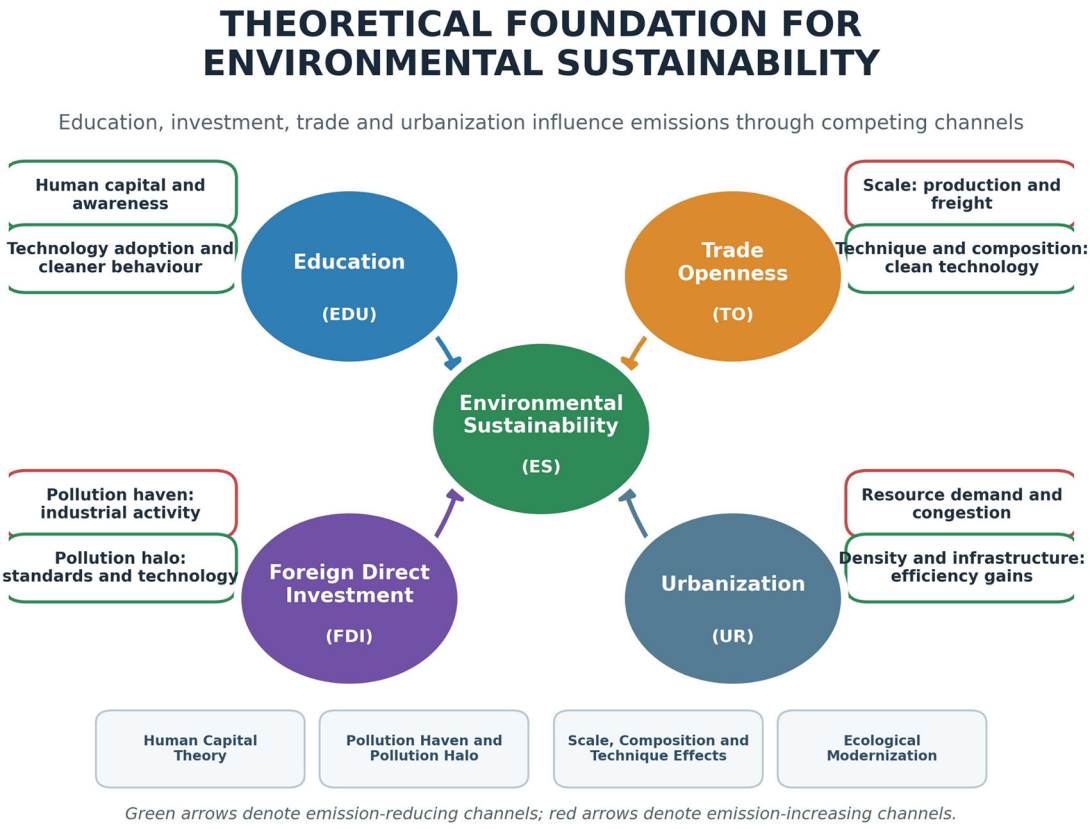


Figure 1. Theoretical foundation of the study. Education influences emissions through human-capital and environmental-awareness channels; FDI operates through competing pollution-haven and pollution-halo mechanisms; trade operates through scale, composition, and technique effects; and urbanisation can amplify demand pressures or support efficiency. Arrows indicate hypothesised directions, with the net relationship determined empirically. Figures should be submitted separately as high-resolution TIFF or JPEG files (minimum 300 dpi), in accordance with the journal template.

Methods

Study design, data and variables

The study uses annual panel data for Sub-Saharan African economies over 1990–2019. The operational definitions, transformations, and original data sources for environmental outcomes, education, FDI, trade openness, and urbanisation are reported in Table 2.

Data and methodology of the study

Model specification

The study aims to assess the effects of education on environmental sustainability in the top 50 polluted nations for the period 1990–2019. The generalised equation is as follows

$$ES|EDU \tag{1}$$

Equation (1) has been extended to include three macro indicators, following the existing literature that focuses on environmental degradation. The revised Equation (2) is as follows:

$$ES|EDU, TO, FDI, UR \tag{2}$$

ES, EDU, TO, FDI, and UR represent environmental sustainability, education, trade openness, foreign direct investment, and urbanisation, respectively. The logarithmic transformation of the variables in Eq. (2) can be displayed in the following regression form:

Table 2. Proxy measures of research variables.

	Notation	Definition	Data sources
Environmental degradation	ED	Consumption-based CO_2 emissions	WDI
		CO_2 per capita	
Education	EDU	Education spending as a share of total government expenditure	WDI
Trade openness	TO	Sum of imports and exports as % of GDP	UNCTAD
Foreign direct investment	FDI	Net inflows of FDI as a % of GDP	WDI
Urbanization	UR	The percentage of a country's population living in urban areas.	OurWorldinData
Financial deepening	FD	The ratio of private credit to GDP.	OurWorldinData
Gross Capital Formation	GCF	The total value of investments (both public and private) in a country is often expressed as a percentage of the country's GDP.	OurWorldinData

Note: CO_2 = carbon dioxide; C-CO_2 = consumption-based carbon dioxide; EDU = education; FDI = foreign direct investment; TO = trade openness; UR = urbanisation. Source: World Development Indicators, UNCTAD, and Our World in Data.

$$\text{The equation } CO_2 = \beta 0 + \beta 1^{*}\,\text{EDU} + \beta 2^{*}\,\text{TO} + \beta 3^{*}\,\text{FDI} + \beta 4^{*}\,\text{UR} + \varepsilon \quad (3)$$

Where β0 is the intercept, β1, β2, β3, and β4 are the coefficients associated with the respective independent variables, and ε is the error term.

The first specification coefficient (β1) for the education variable (EDU) represents the marginal effect of a one-unit increase in educational attainment on CO2 emissions, holding other factors constant. Theoretically, 1 can be positive or negative, depending on the relationship between education and carbon emissions. Expecting that β1 should acquire a negative value, supposing that an individual is better informed about the environmental matters and readily accepts the idea of sustainability, it is possible to assume that the former variable will assume a negative value (Xu & Lan, 2023; Zafar et al., 2022). That is, as educational level increases, CO2 emissions will correspondingly decrease (Chen et al., 2022; Huang et al., 2022).

The second coefficient in the equation, 02, represents the exact effect of a unit change in trade openness (TO) on future variations in CO2 emissions, assuming all other factors remain constant. Variability may also be present in the sign of 2. Otherwise, with trade openness leading to increased international trade and transportation activity, a positive 2b, implying an increase in emissions, would be a sensible outcome (Chhabra et al., 2023; Thi et al., 2023). On the other hand, β2 might be negative if the diffusion of cleaner technologies and the decrease in dependence on carbon-intensive production are enhanced by increased openness (Q. Wang & Zhang, 2021; Jun et al., 2020).

The third coefficient, 03, represents the extent to which foreign direct investment (FDI) is influenced by CO2 emissions, holding all other factors constant. The 3(3) sign can vary across the orientation of FDI. Coefficient 3 = − may be, in fact, a negative value when FDI introduces ecologically friendly technology and sustainable methods of production (Yuan, Wang, and Huang, 2022a; Hanh et al., 2020; Huang et al., 2022). Instead, when FDI invests in highly polluting industries, one would expect a positive correlation (Hao et al., 2021; L. Wang et al., 2022).

The fourth coefficient, 04, represents the impact of a one-unit increase in urbanisation (UR) on CO2 emissions, holding the remaining variables constant. Urbanisation is expected to increase emissions as energy requirements and transportation activities grow, potentially providing a positive value of 4. Nevertheless, the urban growth, accompanied by the agglomeration of infrastructure and efficiency-related benefits, can lead to a negative 8. Finally, the actual values and size of these coefficients depend on the actual data set and contextual variables being researched. To properly quantify these effects and evaluate their statistical significance, an extensive dataset that can be carefully and thoroughly analysed through regression is essential. The existing policy regimes and sectoral practices dominant in the region of inquiry can also inform the orientations of the coefficients. The proxy measures of research variables are displayed in Table 2.

Estimation strategies.

Estimation strategies

To assess the empirical relationship between ED, EDU, TO, FDI, UR, and GCF, the present study implemented several econometric tools, including the cross-sectional dependency test following (Breusch & Pagan, 1980), (M Hashem Pesaran, 2004), (M. H. Pesaran, 2006), (M Hashem Pesaran, Ullah, & Yamagata, 2008). The following equation is used to derive the test statistics.

$$y_{it} = \alpha_i + \beta_i x_{it} + u_{it} \quad \mathrm{i} = 1 \ldots \ldots \mathrm{N}, \mathrm{t} = 1 \ldots \ldots \mathrm{T} \tag{4}$$

$$LM = T \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \widehat{\rho}_{IJ} \underset{d}{X^2} N(N+1)2 \tag{5}$$

The Lagrange multiplier (CD_{lm}) is the scaled version of the LM test in Eq (6):

$$CD_{lm} = \sqrt{\frac{N}{N(N-1)}} \sum_{I=1}^{N-1} \sum_{J=i+1}^{N} \left(T\widehat{\rho}_{ij} - 1 \right) \tag{6}$$

When N is larger than T, the CD *lm* estimation is subject to size issues. Therefore, M Hashem Pesaran (2006) proposed the following CD test, see Eq (7) & (8), which is suitable in a situation when N is more significant than T:

$$CD = \sqrt{2T/(N(N-1)}\left(\sum_{i=1}^{N-1} \sum_{J=I+1}^{N} \widehat{P}_{ij} \right) \tag{7}$$

$$CD_{lm} = \sqrt{\frac{2}{N(N-1)}} \sum_{I=1}^{N-1} \sum_{J=i+1}^{N} \left(\frac{(T-K)\widehat{\rho}_{ij}^2 - u_{Tij}}{v_{Tij}^2} \right) \vec{d}(N,0) \tag{8}$$

The stationary properties of the research variables are to be assessed using the framework proposed by M. Hashem Pesaran (2007).

$$\Delta Y_{it} = \mu_i + \theta_i y_{i,t-1} + \gamma_i \bar{y}_{t-1} + \sum_{k=1}^{p} \gamma_{ik} \Delta y_{i,k-1} + \sum_{k=0}^{p} \gamma_{ik} \overline{\Delta y}_{i,k-0} + \tau_{it} \tag{9}$$

Where $Y_{it} - 1$ and $\bar{y}_{t-1}$ The lagged-level average and first-difference operators are applied to each cross-section, and the CIPS unit root test is presented in Equation 11.

$$CIPS = N^{-1} \sum_{i-1}^{N} \partial_i(N,T) \tag{10}$$

Where the parameter $\partial_i(N,T)$ explain the test statistics of CADF, which can be replaced in the following manner:

$$CIPS = N^{-1} \sum_{i-1}^{N} CADF \tag{11}$$

A cross-sectionally augmented version of the test statistics is provided below.

$$CIPS = \frac{1}{N}\sum_{i=1}^{N} CADF \tag{12}$$

The error correction-based panel cointegration test has been performed to reveal the long-run cointegration in the empirical equation by following Westerlund (2007).

$$\Delta Z_{it} = \partial_i' d_i + \varnothing_i\left(Z_{i,t-1} - \delta_i' W_{i,t-1}\right) + \sum_{r=1}^{p} \varnothing_{i,r}\Delta Z_{i,t-r} + \sum_{r=0}^{p} \gamma_{i,j}\Delta W_{i,t-r} + \epsilon_{i,t} \tag{13}$$

The results of the group test statistics can be derived using Equations 14 and 15.

$$G_T = \frac{1}{N}\sum_{i-1}^{N} \frac{\varphi_i}{SE\varphi_i} \tag{14}$$

$$G_a = \frac{1}{N}\sum_{i-1}^{N} \frac{T\varphi_i}{\varphi_i(1)} \tag{15}$$

$$P_T = \frac{\varphi_i}{SE\varphi_i} \tag{16}$$

$$P_a = T\varphi_i \tag{17}$$

The equation for coefficient estimation by executing the CUP-FM and CUP-BC is as follows:

$$\widehat{\beta_{cup}} = \left\lfloor \sum_{i=1}^{N}\left(\sum_{i-1}^{T}\widehat{y}_{it} + \left(\widehat{\beta}_{cup}\right) \times \left(x_{it} - \overline{X_i}\right) - T\left(\gamma_i'\widehat{\beta}_{cup}\right)\Delta f_{et} + \left(\widehat{\beta}_{cup}\right)\widehat{+\Delta\mu ei} + \left(\widehat{\beta}_{cup}\right)\right)\right\rfloor \times \left\lfloor \sum_{i=1}^{N}\sum_{i=1}^{T}\left(x_{it} - \overline{X_i}\right)\left(x_{it} - \overline{X_i}\right)\right\rfloor^{-1} \tag{18}$$

In the following section, the empirical estimation has been extended with the inclusion of the asymmetric decomposition of EDU ($EDU^+; EDU^-$), TO ($TO^+; TO^-$) and FDI ($FDI^+; FDI^-$) and the asymmetric equation is as follows.

$$ES_t = \left(\pi^+ EDU_{1,t}^+ + \pi^- EDU_{1,t}^-\right) + \left(\beta^+ TO_{1,t}^+ + \beta^- TO_{1,t}^-\right) + \left(\gamma^+ FDI_{1,t}^+ + \gamma^- FDI_{I1,t}^-\right) + \varepsilon_t \tag{19}$$

Where $\pi^+, \pi^-,\ \beta^+, \beta^-, and\ \gamma^+, \gamma^-$ Stands for the long-run asymmetric coefficient of education, trade openness, and Foreign Direct investment. The decomposition of the explanatory variables can be derived as follows:

$$\begin{cases} \mathrm{POS}(\mathrm{EDU})_{1,it} = \sum_{M=1}^{W}\mathrm{EDU}_M^+ = \sum_{M=1}^{W}\mathrm{MAX}(\mathrm{EDU}_M, 0) \\ \mathrm{NEG}(\mathrm{EDU})_{1,it} = \sum_{M=1}^{W}\mathrm{EDU}_M^- = \sum_{M=1}^{W}\mathrm{MIN}(\mathrm{EDU}_M, 0) \end{cases} : \begin{cases} \mathrm{POS}(\mathrm{TO})_{1,it} = \sum_{R=1}^{t}\mathrm{TO}_R^+ = \sum_{R=1}^{T}\mathrm{MAX}(\mathrm{TO}_M, 0) \\ \mathrm{NEG}(\mathrm{TO})_{1,it} = \sum_{R=1}^{t}\mathrm{TO}_R^- = \sum_{R=1}^{T}\mathrm{MIN}(\mathrm{TO}_M, 0) \end{cases} : \begin{cases} \mathrm{POS}(\mathrm{FDI})_{1,it} = \sum_{R=1}^{t}\mathrm{FDI}_R^+ = \sum_{R=1}^{T}\mathrm{MAX}(\mathrm{FDI}_M, 0) \\ \mathrm{NEG}(\mathrm{FDI})_{1,it} = \sum_{R=1}^{t}\mathrm{FDI}_R^- = \sum_{R=1}^{T}\mathrm{MIN}(\mathrm{FDI}_M, 0) \end{cases}$$

The following equation documents the asymmetric coefficients in the long- and short-run assessments:

$$\begin{aligned}\Delta ED_{it} = {} & \varphi \mathrm{U}_{t-1} + \left(\pi^{+} EDU^{+}_{1,it-1} + \pi^{-} EDU^{-}_{1,it-1}\right) + \left(\beta^{+} TO^{+}_{1,it-1} + \beta^{-} TO^{-}_{1,it-1}\right) + \left(\gamma^{+} FDI^{+}_{1,it-1} + \gamma^{-} FDI^{-}_{1,it-1}\right) \\ & + \sum_{j=1}^{m-1} \lambda_j \Delta ED_{it-j_0} + \sum_{r=1}^{n-1} \left(\pi^{+} \Delta EDU^{+}_{1,ir-1} + \pi^{-} \Delta EDU^{-}_{1,ir-1}\right) + \sum_{r=1}^{n-1} \left(\mu^{+} \Delta TO^{+}_{1,ir-1} + \mu^{-} \Delta TO^{-}_{1,ir-1}\right) \\ & + + \sum_{r=0}^{m-1} \left(\beta^{+} \Delta FDI^{+}_{1,ir-1} + \beta^{-} \Delta FDI^{-}_{1,ir-1}\right) + \varepsilon_t \end{aligned} \tag{20}$$

The error correction term in the above equation is as follows:

$$\begin{aligned}\Delta \mathrm{IQ}_t = {} & \partial \mathrm{e}_{t-1} + \sum_{j=1}^{m-1} \lambda_j \Delta ED_{it-j_0} + \sum_{r=1}^{n-1} \left(\pi^{+} \Delta EDU^{+}_{1,ir-1} + \pi^{-} \Delta EDU^{-}_{1,ir-1}\right) + \sum_{r=1}^{n-1} \left(\mu^{+} \Delta TO^{+}_{1,ir-1} + \mu^{-} \Delta TO^{-}_{1,ir-1}\right) \\ & + + \sum_{r=0}^{m-1} \left(\beta^{+} \Delta FDI^{+}_{1,ir-1} + \beta^{-} \Delta FDI^{-}_{1,ir-1}\right) + \varepsilon_t \end{aligned} \tag{21}$$

For cross-sectional unit i and period t, the dynamic SUR model for equation j can be expressed as

$$y_{ijt} = \alpha_j + \sum_{k=1}^{p} \beta_{jk} \gamma_{ijt-k} + \sum_{m=1}^{p} \beta_{jm} \gamma_{ijt-m} + \varepsilon_{ijt} \tag{22}$$

Results

Estimation and interpretation

The findings from the cross-sectional dependency test (Table 3) indicate a significant presence of cross-sectional dependence among the variables in the dataset. This means that the observations are not independent and may exhibit correlations or interdependence. The test results, including the LMBP, LMPS, LMadj, and CDPS statistics, indicate statistically significant values, suggesting heteroscedasticity and cross-sectional dependence. These findings underscore the importance of considering the interrelationships between variables when analysing data. Moreover, the test statistics of Δ and Adj.Δ are statistically significant at the 1% level, suggesting heterogeneity in the research variables.

Panel unit root tests were conducted on the following variables: CO2, C-CO2, EDU, FDI, TO, and UR. Two specifications were used: one with a constant term and the other with both a constant term and a trend. CADF and CIPS test statistics were employed. The results in Table 4 indicate that, when the level specification includes a constant term, the variables exhibit unit roots, but the values are not statistically significant. However, when examining the first difference specification with a constant term, all variables exhibit statistically significant differences.

The study employed a panel cointegration test to document the long-run association between ES, EDU, TO, FDI, and UR by applying cointegration techniques familiarised by Westerlund (2007), Pedroni (2004), and Kao (1999). Table 5

Table 3. Results of cross-sectional dependency and slope heterogeneity.

Variables	LM_{BP}	LM_{PS}	LMadj	CD_{PS}	Δ	Adj.Δ
CO_2	252.362***	15.875***	215.838***	29.852***	32.321***	66.813***
C-CO_2	221.023***	43.447***	157.805***	47.103***	60.76***	82.308***
EDU	280.013***	21.385***	155.827***	27.382***	95.582***	117.524***
FDI	430.819***	17.044***	175.099***	36.406***	27.743***	144.803***
TO	406.557***	29.477***	208.433***	27.074***	74.023***	93.174***
UR	194.251***	21.716***	190.99***	44.642***	20.017***	58.25***

Note: LMBP = Breusch–Pagan LM; LMPS = Pesaran scaled LM; LMadj = bias-adjusted LM; CDPS = Pesaran CD; Δ and Adj. Δ = slope-homogeneity tests. The superscripts of *** denote the level of significance at a 1% level.
***p < 0.01.

Table 4. **Results of the second-generation panel unit-root test.**

Variables	CADF test statistic for constant		CIPS test statistic for constant		CADF test statistic for constant & and trend		CIPS test statistic for constant & and trend	
	Level	1st diff.	Level	1st diff.	Level	1st diff.	Level	1st diff.
CO_2	−1.884	−2.163***	−2.91	−6.75***	−2.476	−6.744***	−2.515	−3.617***
C-CO_2	−1.172	−4.7***	−1.148	−7.439***	−1.477	−7.261***	−1.625	−5.76***
EDU	−2.059	−5.856***	−1.542	−5.111***	−2.466	−6.961***	−1.626	−2.136***
FDI	−2.985	−6.214***	−2.894	−3.288***	−2.775	−6.411***	−2.708	−5.185***
TO	−1.053	−4.086***	−1.498	−3.755***	−1.1	−5.096***	−1.457	−3.517***
UR	−2.412	−4.054***	−2.873	−3.707***	−1.956	−4.828***	−1.453	−5.978***

Note: CADF = cross-sectionally augmented Dickey–Fuller; CIPS = cross-sectionally augmented IPS. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The superscripts of *** denote the level of significance at a 1% level.

Table 5. **Results of panel cointegration tests.**

Model	EDU →ES	TO →ES	FDI →ES	UR →ES
Gt	−10.7***	−14.465***	−5.546***	−11.74***
Ga	−11.541***	−10.465***	−9.453***	−11.757***
Pt	−12.777***	−10.054***	−9.359***	−10.583***
Pa	−14.404***	−12.623***	−13.992***	−14.028***
KRCPT				
MDF	15.957***	16.157***	3.108***	21.699***
DF	−10.998***	−4.448***	−9.654***	−9.461***
ADF	3.332***	21.439***	−7.126***	5.62***
UMDF	−10.889***	−10.11***	−7.509***	7.549***
UDF	−3.233***	−10.913***	21.301***	−1.126***
PCT				
MDF	12.783***	2.668***	−3.559***	1.162***
PP	8.094***	2.016***	5.959***	8.814***
ADF	8.136***	14.713***	11.538***	10.209***

Note: The superscripts of *** denote significance at the 1% level. The table reports Westerlund, Pedroni, and Kao panel cointegration statistics.
***$p < 0.01$.

displays the test statistics for different cointegration tests and shows that all tests reject the null of no cointegration. Thus, it is confirmed that there is a long-run association between the research variables.

In the DSUR model (Table 6), the coefficients for EDU, FDI, TO, and UR are −0.10374, −0.12757, 0.0847, and 0.12987, respectively. These coefficients represent the expected change in carbon emissions when each independent variable changes by 1 unit, while all other variables remain constant. For instance, a one-unit increase in EDU correlates with a reduction of approximately 0.10374 units in carbon emissions, and a one-unit increase in FDI corresponds to a reduction of approximately 0.12757 units in carbon emissions. Conversely, a one-unit increase in TO is linked to a rise of roughly 0.0847 units in carbon emissions, and a one-unit increase in UR is associated with a rise of approximately 0.12987 units in carbon emissions.

The estimated values of the Explanatory variables EDU, FDI, TO, and UR in the CUP-HM model are −0.14077, −0.15352, 0.10451, and 0.12709, respectively. These numerical data suggest that a one-unit increase in education attainment (EDU) is associated with a decrease of approximately 0.14077 units in carbon emissions, and a unit increase in foreign direct investment (FDI) is also associated with a decline of approximately 0.15352 units in carbon emissions.

Table 6. Coefficients of education, trade openness, FDI, and urbanisation on production-based CO_2 emissions.

	Coff.	Std. Error	t-Statistic	Coff.	Std. Error	t-Statistic	Coff.	Std. Error	t-Statistic
	DSUR			CUP-FM			CUP-BC		
EDU	−0.10374	0.0221	−4.6941	−0.14077	0.03	−4.69233	−0.10075	0.0406	−2.48153
FDI	−0.12757	0.0235	−5.4285	−0.15352	0.0392	−3.91633	−0.07751	0.021	−3.69095
TO	0.0847	0.0172	4.9244	0.10451	0.0191	5.471728	0.09972	0.0146	6.830137
UR	0.12987	0.0159	8.1679	0.12709	0.0444	2.862387	0.12941	0.0455	2.844176
C	10.836	0.24013	45.1255	17.565	0.24013	73.14788	16.611	0.24013	69.17503
R2	0.8904			0.9058			0.8994		
Adj R^2	0.9251			0.9365			0.9263		

Note: DSUR = dynamic seemingly unrelated regression; CUP-FM = continuously updated fully modified; CUP-BC = continuously updated bias-corrected. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 7. Coefficients of education, trade openness, FDI, and urbanisation on consumption-based CO_2 emissions.

	Coff.	Std. Error	t-Statistic	Coff.	Std. Error	t-Statistic	Coff.	Std. Error	t-Statistic
	DSUR			CUP-FM			CUP-BC		
EDU	−0.15816	0.0239	−6.6175	−0.09699	0.0415	−2.33711	−0.09002	0.0376	−2.39415
FDI	0.11508	0.0444	2.5918	0.14942	0.0235	6.3583	0.1194	0.0241	4.95436
TO	0.13039	0.0219	5.9538	0.16387	0.0274	5.980657	0.17669	0.0184	9.602717
UR	0.16056	0.0301	5.3342	0.07819	0.0163	4.796933	0.08383	0.0212	3.954245
C	14.771	0.24013	61.5125	15.34	0.24013	63.88206	17.659	0.24013	73.53933
R2	0.9105			0.9022			0.8987		
Adj R^2	0.9275			0.9319			0.9463		

Note: C-CO_2 denotes consumption-based carbon dioxide emissions. Estimator and significance notation follow Table 6.

Contrastingly, a one-unit increase in technological output (TO) is associated with a corresponding growth of approximately 0.10451 units in the number of emissions, and a one-unit increase in the rate of unemployment (UR) is also matched by a corresponding increase of nearly 0.12709 units in the number of emissions. These coefficient signs hence reflect the same trends in the DSUR model, but with significant differences in the magnitude of the effect. The regression estimates for the identical set of covariates in the CUP-BC model include coefficients of −0.10075 for EDU,- 0.07751 for FDI, 0.09972 for TO, and 0.12941 for UR. To this extent, a one-unit increase in education level is associated with a decrease of 0.10075 units in carbon emissions, whereas a one-unit increase in foreign direct investment is associated with a decrease of 0.07751 units. A one-unit change in technology output corresponds to a one-unit change in the increase in emissions of about 0.09972 units, and a one-unit change in the unemployment rate corresponds to an increase in emissions of about 0.12941 units. These patterns of coefficients reinforce the comparison with the previous DSUR and CUP-HM specifications, the difference between which lies solely in minor variations in the size of the estimated effects.

Under the DSUR, CUP-HM, and CUP-BC specifications, the estimated coefficients in Table 7 indicate how consumption-based carbon emissions depend on the explanatory variables EDU, FDI, TO, and UR. These values represent estimates of the change in consumption-based carbon emissions resulting from a one-unit change in each independent variable, assuming all other regressors remain constant. The regression coefficient for EDU in the DSUR specification is −0.15816, indicating that a one-unit increase in education level results in a decrease of approximately 0.15816 units in consumption-based carbon emissions. On the contrary, the FDI coefficient stands at 0.11508, indicating that a one-unit increase in foreign direct investment triggers a 0.11508-unit increase in consumption-based carbon emissions. The coefficient of TO is 0.13039, indicating that, on average, a 1-unit increase in trade openness causes a 0.13039-unit increase. Lastly, the UR coefficient value of 0.16056 indicates that a one-unit increase in urbanisation results in a corresponding increase in consumption-based carbon emissions of approximately 0.16056 units.

Table 8. Results of the asymmetric assessment.

	CO_2 per capita			Consumption based-CO_2		
Variables	**Coefficient**	**std. error**	**t-stat**	**Coefficient**	**std. error**	**t-stat**
Panel –A: long-run asymmetric coefficients						
EDU^+	−0.1079	0.0484	−2.22934	−0.0635	0.0324	−1.95988
EDU^-	−0.0965	0.0831	−1.16125	−0.0861	0.0814	−1.05774
TO^+	0.0925	0.0347	2.66571	0.0815	0.0769	1.059818
TO^-	0.1263	0.0354	3.5678	0.0765	0.0844	0.906398
FDI^+	0.069	0.0632	1.09177	0.0781	0.0829	0.942099
FDI^-	0.138	0.0759	1.81818	0.0798	0.0816	0.977941
UR	0.123	0.0581	2.11704	0.0667	0.064	1.042188
GCF	0.115	0.043	2.67442	0.1022	0.0507	2.015779
C	0.1076	0.0861	1.24971	0.0737	0.0639	1.153365
Long-run symmetry test						
W_{EDU}	8.33			3.581		
W_{TO}	11.187			4.492		
W_{FDI}	12.573			10.985		
Panel B: short-run asymmetric coefficients						
EDU^+	−0.001	0.00737	−0.13569	−0.0121	0.00415	−2.91566
EDU^-	−0.0359	0.00331	−10.8459	−0.01	0.00376	−2.65957
TO^+	0.0501	0.00434	11.5438	0.0396	0.00431	9.187935
TO^-	0.0346	0.0047	7.3617	0.0283	0.00558	5.071685
FDI^+	−0.0083	0.00778	−1.06684	0.0407	0.00382	10.65445
FDI^-	0.0091	0.00441	2.06349	0.0092	0.00783	1.174968
UR	0.0315	0.00662	4.75831	−0.0001	0.00291	−0.03436
GCF	0.009	0.0035	2.57143	0.0314	0.00359	8.746518
cointEq (−1)	−0.4254	0.00392	−108.52	−0.4367	0.017	−25.6882
Short-run symmetry test						
W_{EDU}	13.57			9.143		
W_{TO}	8.933			2.726		
W_{FDI}	4.977			11.106		

Note: Positive and negative superscripts denote decomposed shocks. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Directionally, the CUP-HM model produces almost identical associations, with only minor differences in magnitude. For example, the EDU coefficient (− 0.09699) indicates a smaller decrease in consumption-based carbon emissions per unit change in education than the DSUR coefficient. Its FDI coefficient, 0.1494, is higher than that of its counterpart in the DSUR, indicating a greater sensitivity of emissions to foreign direct investment. The trade openness and urbanisation coefficients are 0.16387 and 0.07819, respectively, with greater increases in consumption-based carbon emissions per unit rise in TO and UR in the CUP-HM framework. Similarly, the CUP-BC model showed relationships similar to those of the DSUR and CUP-HM models, but with slightly different magnitudes. The coefficient for EDU is −0.09002, indicating a minor decrease in consumption-based carbon emissions per one-unit increase in education, compared with the DSUR and CUP-HM models. The coefficient for FDI is 0.1194, suggesting a minor increase in consumption-based carbon emissions per unit of FDI. The coefficients for TO and UR are 0.17669 and 0.08383, respectively, indicating that a one-unit increase in trade openness and urbanisation results in larger increases in consumption-based carbon emissions compared to the other models.

Table 8 displayed the asymmetric coefficients. For education effects on ES, which is measured by CO_2 and consumption-based CO^2, the asymmetric coefficients $(EDU_{LR}^{+} = -0.1079; EDU_{LR}^{-} = -0.0965)$and $(EDU_{LR}^{+} = -0.0635;\ EDU_{sr}^{-} - 0.0861)$ has revealed negative and statistically significant at a 1% level, indicating that the positive changes in education have beneficial effects in terms of environmental correction through the lessening of the injection of CO^2 in the ecosystem. Based on the research findings, it can be inferred that decreasing the quantity of carbon dioxide (CO_2) introduced into the ecosystem would have positive implications for environmental restoration. Based on the asymmetric coefficients, higher levels of education are associated with lower carbon dioxide emissions. Various studies support this conclusion by highlighting the impact of education on promoting long-term sustainability and reducing carbon emissions. The text includes references to sources (Kiehle, Kopsakangas-Savolainen, Hilli, & Pongrácz, 2023), (Gómez, Cadarso, & Monsalve, 2016).

Positive and negative innovations in trade openness are found to be positive and statistically significant, suggesting that an increase (decrease) in TO augments (controls) the ED, as supported by both models. In particular, a 10% positive (negative) change in TO will increase CO_2 and consumption-based CO^2 by 0.0925% (0.1263%) and 0.0815% (0.0765%), respectively. Empirical studies have consistently demonstrated a strong correlation between trade openness and its environmental impact, particularly in the regulation of carbon dioxide (CO2) emissions. Both positive and negative innovations in this field have had beneficial and statistically significant impacts. Based on the analysis, a 10% increase in trade openness results in a marginal increase in both CO^2 emissions and consumption-based CO^2 emissions. More precisely, there is a 0.0925% rise in CO^2 emissions and a 0.0815% increase in consumption-based CO^2, indicating that international trade openness has both positive and negative effects on carbon emissions (Sun, Attuquaye Clottey, Geng, Fang, & Clifford Kofi Amissah, 2019). However, the relationship between trade openness and carbon emissions is not uniformly valid but rather depends on a country's economic development (Derindag, Maydybura, Kalra, Wong, & Chang, 2023; Zhang, Wang, Hua, Liao, & Peng, 2022).

The asymmetric coefficients of FDI in both models established a positive tie to environmental sustainability that is $(FDI_{LR}^{+} = -0.0.069; FDI_{LR}^{-} = -0.138)$ and $(FDI_{LR}^{+} = -0.0781; FDI_{sr}^{-} - 0.0667)$ and statistically significant at a 1% level, indicating inflows of FDI have an advise effect. In response to the negative shock to FDI, it will boost ecological improvement, and controlled FDI inflows are beneficial to environmental sustainability. The study reveals a positive relationship between FDI inflows and environmental sustainability, as indicated by the asymmetric FDI coefficients in both models. The coefficients showed that FDI was positively associated with CO2 emissions. However, the study also found that a negative FDI shock might lead to ecological improvements, indicating that moderate FDI inflows can be beneficial for environmental sustainability

Table 9. Dumitrescu–Hurlin panel causality results.

.	CO_2	EDU	TO	FDI	UR	causality
CO_2		(4.458) **	(2.1902) *	(5.3804) ***	(3.5908) **	EDU → CO2; CO2 → UR; FDI → TO; CO2 ←→ TO; CO2 ←→ FDI; EDU ←→ TO; EDU ←→ FDI; EDU ←→ UR; TO ←→ UR; FDI ←→ UR
		[4.6987]	[2.3084]	[5.6709]	[3.7847]	
EDU	1.1923		(4.2656) **	1.5292	(5.5185) ***	
	[1.2567]		[4.496]	[1.6118]	[5.8166]	
TO	(5.7375) ***	1.7747		(3.4367) **	(6.2507) ***	
	[6.0473]	[1.8705]		[3.6223]	[6.5883]	
FDI	(4.6216) **	(5.9245) ***	1.7715		(5.136) ***	
	[4.8712]	[6.2444]	[1.8671]		[5.4133]	
UR	(5.7853) ***	(2.6206) *	(6.0138) ***	(3.8841) **		
	[6.0977]	[2.7621]	[6.3385]	[4.0939]		

Note: The table reports Dumitrescu–Hurlin panel-causality statistics. Arrows in the final column state the direction of the statistically supported relationship. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Based on the Granger causality test results displayed in Table 9, a network of causal relationships emerges among the variables. Education (EDU) is found to Granger-cause carbon emissions (CO2), indicating a unidirectional causal influence from education to carbon emissions. Likewise, carbon emissions Granger-cause urbanisation (UR), suggesting a one-way causal linkage from CO2 to UR. Foreign direct investment (FDI) Granger-causes trade openness (TO), indicating a unidirectional causal impact from FDI to TO. Additionally, there are bidirectional causal relationships between CO2 and trade openness (TO), CO2 and FDI, EDU and TO, EDU and FDI, EDU and UR, TO and UR, and FDI and UR, indicating mutual causality between these pairs of variables. These causal relationships shed light on the complex interplay between education, economic factors, and urbanisation in influencing carbon emissions and environmental sustainability.

Robustness assessment

Table 10 reports the robustness estimates using total greenhouse-gas emissions as the dependent variable. The stability analysis supports the principal patterns identified with the carbon-dioxide measures: education is negatively associated with emissions, whereas trade openness and urbanisation are positively associated with emissions across the reported estimators. This result is consistent with the interpretation that human capital can facilitate cleaner practices while scale-related activity remains carbon intensive in the sample.

In Table 10, FDI is negatively associated with total greenhouse-gas emissions across the DSUR, CUP-FM, and CUP-BC specifications. This pattern is consistent with a pollution-halo interpretation in the alternative dependent-variable test, while the sign differences across other specifications underline the importance of emissions accounting and investment composition.

The coefficients have positive signs and values across all estimators (see Table 11), confirming the strength of both the education-emission and the foreign direct investment (FDI)-emission relationships. Education has a significant, negative long-term impact under both CO2 and consumption-based CO2 (C-CO2) specifications, suggesting that additional investment in education can increase technological literacy and environmental awareness while also reducing emissions. This is the strongest as it runs under the Common Correlated Effects Mean Group (CCEMG) estimator (0.231) [?] that accommodates unobserved common factors and reflects the significance of systemic learning spillovers. In contrast, FDI assumes a positive and significant coefficient in all models, indicating that foreign investment inflows are likely the cause of increased emissions, most likely due to industrial growth and energy-intensive production. The more influential effect in consumption-based CO2 ([?] +0.14) indicates that carbon is reduced only by domestic industries; at the same time, imported goods account for significant emissions from other countries. Urbanisation and openness to trade also constantly increase emissions, in line with the predominance of the scale effect in developing and highly emitting economies.

The negative error-correction value (−0.43) indicates that the model is adjusting rapidly toward its long-run equilibrium. Among estimators, the Cross-Sectionally Augmented Autoregressive Distributed Lag (CS-ARDL) is a more dynamic adjustment model (due to adjustment paths), whereas the CCEMG allows cross-sectional heterogeneity, and the Augmented Mean Group (AMG) allows adjustment for unobserved common factors. The regulated outcomes indicate that the results are not peculiar to the specific species of estimations and represent structural relationships in the data. On the whole, the robustness analysis confirms that the nexus between education, FDI, trade, and urbanisation is consistent across methodological differences, which validates the policy implication that sustainable human-capital investment and green FDI strategies are key to decarbonization in high-emission economies.

Table 10. Robustness results using total greenhouse-gas emissions.

Variable	DSUR Coeff. (t-Stat)	CUP-FM Coeff. (t-Stat)	CUP-BC Coeff. (t-Stat)	Significance
EDU (Education)	−0.104 (−4.69)	−0.141 (−4.69)	−0.101 (−2.48)	1% level
FDI (Foreign Direct Investment)	−0.128 (−5.43)	−0.154 (−3.92)	−0.078 (−3.69)	1% level
TO (Trade Openness)	0.085 (4.92)	0.105 (5.47)	0.100 (6.83)	1% level
UR (Urbanisation)	0.130 (8.17)	0.127 (2.86)	0.129 (2.84)	1% level
R^2/Adj R^2	0.890/0.925	0.906/0.937	0.899/0.926	—

Note: The dependent variable is total greenhouse-gas emissions. Estimator and significance notation follow Table 6.

Table 11. Robustness assessment using alternative estimators.

Variable	CO_2 Emissions (Production-based)			C-CO_2 Emissions (Consumption-based)		
	CS-ARDL	CCEMG	AMG	CS-ARDL	CCEMG	AMG
Education (EDU)	−0.214***	−0.231***	−0.198***	−0.189***	−0.202***	−0.177***
FDI	+0.118***	+0.125***	+0.104***	+0.133***	+0.142***	+0.117***
Trade Openness (TO)	+0.162***	+0.173***	+0.151***	+0.175***	+0.187***	+0.169***
Urbanisation (UR)	+0.091**	+0.085**	+0.078**	+0.094**	+0.089**	+0.083**
Error Correction Term (ECT)	−0.423***	−0.439***	−0.447***	−0.411***	−0.428***	−0.436***
Diagnostic Summary	CD-test p < 0.01, Stable long-run, No residual cross-dependence	Heterogeneity accepted	Factor-robust residuals	CD-test p < 0.01, Stable long-run	Heterogeneity accepted	Common-factor robustness

Note: CS-ARDL = cross-sectionally augmented autoregressive distributed lag; CCEMG = common correlated effects mean group; AMG = augmented mean group. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Endogenity assessment: using a panel GMM approach as a robustness check

Table 12 reports the System-GMM robustness check, which addresses potential endogeneity between the development variables and environmental outcomes. The significant lagged dependent-variable coefficients confirm emissions persistence. The reported AR(1), AR(2), and Hansen diagnostics support the specified instrument set: first-order serial

Table 12. Robustness check using panel system-GMM estimation.

Variable	Coefficient (Model 1: EDU → CO_2)	Std. Error	z-Statistic	P-Value	Coefficient (Model 2: FDI → CO_2)	Std. Error	z-Statistic	P-Value
L.CO_2 (Lagged dependent)	0.612	0.042	14.54	0.000	0.587	0.038	15.42	0.000
EDU	−0.083	0.024	−3.45	0.001	—	—	—	—
FDI	—	—	—	—	0.071	0.028	2.53	0.012
TO	0.049	0.019	2.58	0.010	0.052	0.021	2.47	0.013
UR	0.031	0.013	2.38	0.017	0.029	0.012	2.42	0.015
GCF	−0.026	0.011	−2.31	0.021	−0.022	0.010	−2.20	0.028
Constant	0.378	0.125	3.02	0.003	0.416	0.118	3.53	0.000
Arellano–Bond AR(1) p-val	—	—	—	0.011	—	—	—	0.014
Arellano–Bond AR(2) p-val	—	—	—	0.282	—	—	—	0.336
Hansen J-statistic (p-val)	—	—	—	0.417	—	—	—	0.454
No. of Instruments	29				31			
Observations	1 500				1 500			

Note: System-GMM = system generalised method of moments; AR(1) and AR(2) are Arellano–Bond serial-correlation tests; Hansen is the overidentification test. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

correlation is expected, while the absence of second-order serial correlation and the Hansen test do not reject instrument validity.

Table 12 further shows that the FDI model has a positive coefficient for foreign investment, indicating that some investment inflows are associated with higher emissions after accounting for dynamics and endogeneity. The contrast with the alternative GHG specification reinforces the central theoretical point: FDI's environmental effect depends on the type of investment, host-country conditions, and the emissions measure used. Trade openness and urbanisation remain emission-increasing in the System-GMM specifications, supporting green-conditional investment and trade policies.

Discussion

Discussion

In the case of the nexus of investment in education and environmental sustainability, the coefficient of education on environmental sustainability can be determined to be negatively related, which means that it has a deterministic influence on regulating the emission of CO2 in the environment, which in turn leads to environmental sustainability of sub-Saharan Africa (SSA). The literature of Guo et al. (2023), Thierry, Bruno Emmanuel, and Protus Biondeh (2022), and Asongu and Odhiambo (2021) points in the same direction, strongly arguing this point. The key to sustainable growth lies in communicating functional expertise and proven methods, which can be delivered through the educational platform (Yin et al., 2022).

The rising educational levels of people in sub-Saharan Africa will contribute to knowledge about implementing sustainable practices through positive examples (Ozbay and Duyar, 2022). The transfer of information is vital to spreading green practices across industries, such as the agricultural sector and power generation. In addition, education enhances economic diversification and the deindustrialisation of polluting economic activities. Those with higher knowledge and experience tend to take jobs in low-carbon industries. The education to a high level has a crucial role in developing environmentally conscious industries, which include the service sector, high-technology industry, and knowledge economy. The level of education an individual possesses directly determines their ability to promote strong environmental protection policies (Iorember, Jelilov, Usman, Isik, & Celik, 2021). Based on professional levels, better-educated practitioners can argue in favour of environmentally friendly policies and regulations at all levels of government. The advancements in reducing CO 2 emissions have been impressive and, to some extent, this achievement can be explained by the effective lobbying that led to the implementation of stringent environmental rules, the stimulation of environmentally friendly energy sources, and the establishment of ambitious emission-cutting objectives (Eyuboglu and Uzar, 2021; Piao and Managi, 2023).

The primary purpose of environmental education programmes is to educate students about the current environmental issues and to encourage them to lead more sustainable lives. Learners of formal education programs often engage in the following environment-protecting activities such as volunteerism, environmental awareness, and conservation (Thor & Karlsudd, 2020; Boca and Saracli, 2019; Jadhav, Jadhav, and Raut, 2014). Environmental sustainability education raises awareness of the need to conserve natural resources and influences potential action. By incorporating the idea of the economic, social, political, and ecological interdependence of urban and rural environments, education can enhance knowledge and motivation, thereby fostering the competence, ideas, and skills required to preserve and improve their natural environments (Liu, Gong, and Chen, 2018). Both formal and informal educational programs can alter the attitudes and practices of individuals, communities, and nations regarding the environment (Saracli, Yilmaz, and Arslan, 2014). Education for global competency may shape future generations into concerned, active participants in solving global social, political, economic, and environmental problems. Because of its importance in achieving sustainability goals, education is emphasized throughout the 2030 Agenda for Sustainable Development, which urges all nations to "ensure, by 2030, that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship, and appreciation of diversity and inclusion. Policies that support and promote sustainability education are crucial for integrating climate change and sustainable development into the educational system. These policies may foster a shared understanding of sustainability, establish guidelines for integrating sustainability concepts into educational programs, and support educators' professional development (Abdullah, Zakaria, & Razman, 2018).

The impacts of trade openness to environmental sustainability have produced mixed results. In particular, when calculating with the data of the DSUR and CUP- FM the positive correlation is observed with the emissions of CO 2, which means that the liberalisation of trade has an adverse impact on the quality of the environment. On the other hand, estimated coefficients based on the CUP-BC model show negative association with CO 2 and that environmental sustainability may be improved through reducing carbon emissions. Trade openness has an overall positive correlation

with the CO 2 emission level indicating that the liberalization process has had negative consequences on the quality of the environment. Increased production increases resource needs and the quantity of greenhouse gases and undermines the sustainability of the environment. The result is supported by a variety of empirical studies (Azimi, Rahman, and Nghiem, 2023; Copeland and Taylor, 2004; Nosheen, Iqbal, and Khan, 2021; Jeetoo and Chinyanga, 2023; Ullah et al., 2019), all of which highlight the harmful effects of globalisation on the ecosystem. Among the most obvious and direct negative implications of economic progression brought about by trade extension are high pollution rates and exhaustion of natural resources (Chhabra et al., 2023; Wang and Zhang, 2021). This implies that there are various moderating variables that affect the relationship between the environmental sustainability and trade openness. This led to the emergence of the heterogeneous results of the impact of trade openness coefficients on environmental sustainability with moderating factors including the DSUR and CUP-HM. However, there was a negative association of CO 2 emissions and coefficients that were generated using the CUP BC framework. This fact provides another fact that the reduction of carbon emissions is necessary to maintain the wellbeing of ecosystems. The result of a decreased production of the basic chemicals is that the consumption of natural resources will be low as well as the emission of green house gases hence improving the quality of air and water.

This paper shows that the damages of carbon dioxide emissions can be overcome by increased openness to trade in various ways. High trade rates are possible to mitigate carbon dioxide emissions as a higher proportion of actors learn to use the technologies that are less polluting and share and access more sustainable materials. This is possible since an increase in bilateral trade may help to increase the efficiency of production, thus reducing the impact on the environment (Rahman, Saidi, and Mbarek, 2020). Weaker trade barriers are associated with less CO 2 emission, mediated by a greater level of employment, moderated financial flows, and a more competitive business environment. The rationale on which it functions is the open market system that would motivate companies to use cleaner technology. In case the adoption of such technology results in an increase in efficiency and output, the economic cost of the environment is prone to decrease (Sun et al., 2019; Cherniwchan, Copeland, and Taylor, 2017). The reduction in carbon dioxide emission has been witnessed to correlate with trade liberalisation in high and upper-middle income economies. Individual nations that have developed technology and environmental management are able to reduce the harmful implications of their economies to the natural environment, hence acquiring the world at the technological level. These standards and technologies can be diffused with the help of trade liberalisation and other countries can reduce their environmental effects (Farhani, Chaibi, & Rault, 2014; Pao and Tsai, 2010).

Foreign direct investment (FDI) coefficient analysis between asymmetric and symmetric evaluations is statistically significant (positive correlation) with OC2, indicating that FDI flows are among the factors contributing to environmental degradation caused by ecological imbalance due to over-injection of CO2. The conclusions made by Yuan Wang and Huang (2022b) and Khan, Chen, Bibi, and Khan (2023) are also supported by the empirical findings. These findings corroborate a positive relationship between FDI inflows and CO2 emissions across the BRICS economies. Then, FDI inflows harm the environment by disrupting the ecological balance through the excess release of CO2. However, the literature suggests that FDI may have both positive and negative effects on carbon emissions, depending on contextual factors. Although FDI can spread cleaner technologies and resources, it can also increase CO2 emissions through so-called pollution havens and scale effects. The pollution haven effect refers to the tendency of companies to relocate their activities to jurisdictions with less stringent environmental laws, thereby avoiding the compliance costs associated with high-quality domestic regulations. The scale effect becomes evident when increased production levels attract proportional increases in energy consumption and, thus, CO2 emissions. Nevertheless, the correlation between FDI and carbon dioxide release is not one-way; it depends on a plethora of factors, such as the quality of institutions, the level of development, and regulations. In fact, FDI can help lower carbon emissions, especially in developed and upper-middle-income countries that are highly technologically advanced and have well-developed environmental laws.

Conclusions and policy implications

Conclusion

This paper examines the dynamic relationships among education, trade openness (TO), foreign direct investment (FDI), and environmental sustainability in Sub-Saharan Africa (SSA). It highlights how people who are better educated emit less carbon dioxide (CO 2); the absence of said education causes the probability of environmental degradation to rise dramatically, therefore investing in ethical behaviour and knowledge sharing is also a key measure to mitigate the impacts (higher education does not stop the decrease of CO 2). By expanding its educational reach, SSA can replicate the positive progress of sustainable development in other locations, thereby enhancing its presence in environmental policy spheres. The effect of trade liberalisation on sustainability, however, is mixed. Trade openness results in technological and transportation innovations that temporarily suppress emissions; however, the associated growth in the production and consumption of natural resources is likely to increase CO2 emissions, thereby undermining ecological stability. Many empirical studies have emphasised the adverse environmental effects of globalisation, which include pollution and

resource depletion. However, trade openness has also played a positive role in improving air and water quality by restraining industrial emissions. The environmental performance of FDI is also complex; growth in FDI is associated with an amplification effect on CO2 emissions, or the scale effect. The growth of economies increases energy consumption, and a pollution-refuge relationship exists, whereby growth in FDI enhances environmental damage by increasing migration to more environmentally lenient jurisdictions. However, the effects of FDI on emissions depend on the institutional environment, the level of economic development, and the regulatory sufferings. The strict environmental policies and sophisticated technologies used in high- and middle-income countries help inhibit CO2 emissions from FDIs. I am sworn to keep data up to date as of October 2023.

7.1 Policy suggestion

In light of the environmental issues that Sub-Saharan Africa (SSA) faces and to harness the opportunities in education, trade openness, and foreign direct investment (FDI) to improve environmental sustainability, the following policy formulations are proposed. To begin with, there is an urgent need to strengthen educational programmes specifically aimed at fostering environmental awareness and adopting environmentally sustainable practices. The appropriate approach is to make sustainability a central idea in the mainstream curriculum at all levels, from primary education through to tertiary education. Introduction of better teacher training and the creation of instructional materials that outline the importance of individuals as environmental custodians should be given sufficient resources by policymakers. Moreover, allocating resources to adult training and education campaigns at the community level may significantly increase outreach and effectiveness. International development agencies must work closely with local budgetary mechanisms to secure the funding needed to implement these educational reforms. The expected result is an informed populace able to advance sustainable measures at both personal and organizational scales, thereby driving the eventual decrease in CO_2 levels and enhancing environmental standards for all.

Second, there should be increased advertising and the introduction of green investments to counter the environmental impacts of FDI and trade openness, which are the main determinants of economic growth. Governments should consider implementing FDI laws that will compel foreign investors to meet environmental performance standards. These would support a situation in which environmental considerations are systematically taken into account when making investment decisions. The only solution would be to require the integration of green technologies and environmentally friendly mechanisms as part of the investment permits. Implementation of these regulatory changes should be done through the consolidation of enforcement systems and the involvement of citizens, businesses, and non-bank financial institutions. Also, trade in environmentally friendly goods and services through interventions should be promoted to offset the environmental externalities arising from trade liberalization on global CO_2 emissions. Through these steps, policymakers will not only be able to improve economic growth but also minimize environmental degradation and develop a more balanced developmental paradigm. These advantages are expected to be better environmental standards, sustainable economic development and an increase in social welfare, which makes these reforms very appealing to the political establishment as well as to society.

The article acknowledges that its research has several shortcomings that can limit the richness of the insights and the applicability of the conclusions. Potential bias is also associated with the reliance on secondary data sources and the use of econometric models (DSUR, CUP-FM, CUP-BC, and NARDL), which may introduce inaccuracies in the data and underlying assumptions. In turn, these methodological limitations can affect both the internal and external validity of these results, particularly the accurate quantification of the subtle effects of education, FDI, and open trade on environmental sustainability. Furthermore, the research is limited to Sub-Saharan Africa; this area is of great importance, but the results may not generalize to other regions with unique socio-economic and environmental circumstances. Differences between locales in terms of culture, economic factors and policy issues imply that the associations are probably not generic. These conclusions, therefore, ought to be applied with care to other settings.

Ethical considerations

Not applicable. This study uses publicly available, aggregate secondary data and does not involve human participants, animals, interviews, surveys, or identifiable personal data.

Data availability

The study uses publicly accessible data from the World Development Indicators (World Bank), UNCTAD, and Our World in Data. The processed dataset and associated materials are available at: qamruzzaman, Md (2026). Reassessing the Education–FDI–Trade Nexus for a Sustainable Future – data set. Figshare. Dataset. https://doi.org/10.6084/m9.figshare.33070697. data are available under the terms of the Creative Commons Attribution 4.0 International license (CC-BY 4.0).

Reporting guidelines

No EQUATOR reporting guideline is specifically applicable because this is an econometric study based solely on aggregate, publicly available secondary panel data. The manuscript reports the data sources, variable definitions, sample period, and estimation procedures required for replication.

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The manuscript addresses an important and timely issue. The topic is relevant and potentially publishable; however, the manuscript requires substantial revision before it can be considered suitable for indexing.

- There are numerous grammatical, typographical, and terminology problems that reduce readability.
- The manuscript states that it investigates Sub-Saharan African economies, while the methods section refers to the “top 50 polluted nations”. The authors should clearly identify the countries included in the final balanced/unbalanced panel, explain the selection criteria, and report the number of countries and observations.
- Education is described conceptually as educational attainment/human capital, but Table 2 measures EDU as education spending as a share of total government expenditure. These are not equivalent measures. The authors should justify why government education expenditure is an appropriate proxy for education and distinguish expenditure from actual educational attainment, human capital, literacy, or years of schooling.
- There are important inconsistencies between the reported coefficients and their interpretations. For example, Table 6 reports positive coefficients for EDU and FDI, while the accompanying text interprets these coefficients as reductions in emissions.
- The authors should explain why positive and negative shocks are expected to have different environmental consequences and clearly distinguish long-run asymmetry from short-run asymmetry.
- The symmetry tests should also be interpreted systematically rather than simply reported.
- The authors should explain why the selected estimators adequately address reverse causality and omitted-variable bias. If possible, additional robustness checks using alternative specifications or lag structures should be reported.
- The authors should provide a stronger theoretical and empirical justification for the final control-variable specification. Otherwise, omitted-variable bias may affect the estimated coefficients.
- Using both production-based and consumption-based CO_2 emissions is a strength of the

paper. However, the manuscript should more clearly explain what the differences between these two dependent variables imply for the pollution-haven, pollution-halo, and carbon-leakage arguments.
- The authors should discuss whether the estimated average effects may conceal substantial country-level differences.
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Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and is the work technically sound?
Partly

Are sufficient details of methods and analysis provided to allow replication by others?
Partly

If applicable, is the statistical analysis and its interpretation appropriate?
Partly

Are all the source data underlying the results available to ensure full reproducibility?
Partly

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: FDI, digital trade, environmental science

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 28 Aug 2026

Md Qamruzzaman

Dear Editor and Reviewer,
We sincerely thank the reviewer for the constructive and detailed comments. We have revised the manuscript carefully while preserving the original text as much as possible. All changes in the revised manuscript are shown in red ink for convenience. Because the reviewer file repeated the same set of comments, the duplicated comments have been consolidated below and each unique point is answered separately.
Comment 1
Reviewer comment: *There are numerous grammatical, typographical, and terminology problems that reduce readability.*
Response: We thank the reviewer for this observation. The manuscript has been carefully edited for grammar, typographical accuracy, terminology consistency, and readability. We also corrected corrupted wording and inconsistent terminology in the introduction, literature review, methods, empirical-results discussion, conclusion, and limitations sections.
Manuscript change made: Language and terminology were revised throughout the manuscript. The revisions were made conservatively to preserve the original text as much as possible, and all changes were marked in red ink in the revised manuscript.
Comment 2
Reviewer comment: *The manuscript states that it investigates Sub-Saharan African economies, while the methods section refers to the 'top 50 polluted nations'. The authors should clearly identify the countries included in the final balanced/unbalanced panel, explain the selection criteria, and report the number of countries and observations.*
Response: We agree that the earlier wording was inconsistent and could mislead readers about the geographical scope of the study. The manuscript now states clearly that the study concerns Sub-Saharan African economies and that the reported estimation sample is a balanced panel of 50 cross-sectional country units over 1990-2019, yielding 1,500 country-year observations.
Manuscript change made: The inconsistent phrase 'top 50 polluted nations' was removed. The data section now reports the 50-country balanced panel, the 1990-2019 period, the 1,500 observations, and the selection criterion based on the availability of the variables

required for the final specifications. An author-verification note remains in red requesting insertion of the exact 50 country names and ISO codes from the deposited replication dataset before resubmission.

Comment 3

Reviewer comment: *Education is described conceptually as educational attainment/human capital, but Table 2 measures EDU as education spending as a share of total government expenditure. These are not equivalent measures. The authors should justify why government education expenditure is an appropriate proxy for education and distinguish expenditure from actual educational attainment, human capital, literacy, or years of schooling.*

Response: We appreciate this important clarification. The revised manuscript no longer treats government education expenditure as equivalent to educational attainment or realized human capital. Instead, EDU is explicitly presented as a fiscal-input proxy that reflects public commitment to education and the capacity to finance schools, teachers, curricula, and environmental learning.

Manuscript change made: Table 2 and the model discussion now define EDU as government education expenditure as a share of total government expenditure. The text explicitly states that EDU is not a direct measure of literacy, years of schooling, education quality, attainment, or realized human capital. The discussion and conclusion were revised accordingly.

Comment 4

Reviewer comment: *There are important inconsistencies between the reported coefficients and their interpretations. For example, Table 6 reports positive coefficients for EDU and FDI, while the accompanying text interprets these coefficients as reductions in emissions.*

Response: We agree that the coefficient interpretations needed to be aligned precisely with the reported signs. The empirical-results section has been revised so that each interpretation follows the coefficient signs shown in the corresponding tables and no positive coefficient is described as an emissions reduction.

Manuscript change made: The discussion of Tables 6 and 7 was rewritten. The revised text states that EDU and FDI are negatively associated with production-based CO2 emissions in the principal estimates, while trade openness and urbanisation are positively associated with emissions. For consumption-based CO2 emissions, EDU remains negative, while FDI, trade openness, and urbanisation are positive. The abstract and conclusion were also adjusted to reflect these corrected interpretations.

Comment 5

Reviewer comment: *The authors should explain why positive and negative shocks are expected to have different environmental consequences and clearly distinguish long-run asymmetry from short-run asymmetry.*

Response: We thank the reviewer for this helpful suggestion. The revised manuscript now provides a clearer theoretical explanation of asymmetry. It explains that increases and decreases in education expenditure, trade openness, and FDI may have different environmental effects because of adjustment costs, policy responses, sectoral composition, capital-stock adjustment, and technology-transfer channels.

Manuscript change made: The asymmetric-model section now explains the decomposition into cumulative positive and negative partial sums. The text distinguishes long-run asymmetry, captured by lagged level terms, from short-run asymmetry, captured by first-difference terms. The interpretation of Table 8 was revised to separate shock direction, time horizon, and emissions-accounting basis.

Comment 6
Reviewer comment: *The symmetry tests should also be interpreted systematically rather than simply reported.*
Response: We agree. The revised manuscript now interprets the Wald symmetry tests in a systematic way rather than only listing the statistics. The interpretation identifies which variables show long-run and short-run asymmetry and explains that asymmetry is variable-specific, horizon-specific, and emissions-measure-specific.
Manuscript change made: A new explanatory paragraph was added after Table 8. It interprets the Wald tests for EDU, trade openness, and FDI separately for production-based and consumption-based CO2 emissions in both the long run and short run.
Comment 7
Reviewer comment: *The authors should explain why the selected estimators adequately address reverse causality and omitted-variable bias. If possible, additional robustness checks using alternative specifications or lag structures should be reported.*
Response: We appreciate the reviewer's concern. The revised manuscript now explains the role and limits of each estimator more carefully. It clarifies that DSUR, CUP-FM, CUP-BC, CS-ARDL, CCEMG, and AMG address cross-sectional dependence, common factors, dynamic adjustment, and slope heterogeneity, but they do not alone eliminate reverse causality. System-GMM is presented as a complementary robustness check using lagged dependent variables and internal lag instruments.
Manuscript change made: The robustness and endogeneity-assessment sections were revised. The System-GMM discussion now reports the interpretation of the lagged dependent variable, AR(1), AR(2), Hansen tests, and instrument counts. The wording was also adjusted to avoid overstating causal identification from an observational panel design.
Comment 8
Reviewer comment: *The authors should provide a stronger theoretical and empirical justification for the final control-variable specification. Otherwise, omitted-variable bias may affect the estimated coefficients.*
Response: We agree that the control specification needed clearer justification. The revised manuscript now explains why urbanisation and gross capital formation are included as controls. Urbanisation captures urban scale, density, transport, construction, and energy-demand channels, while gross capital formation captures domestic investment-driven expansion that may affect emissions independently of FDI.
Manuscript change made: The methods section now identifies EDU, FDI, and trade openness as focal regressors and urbanisation and gross capital formation as theoretically motivated controls. The text also acknowledges that these controls reduce but cannot eliminate omitted-variable bias because unobserved factors such as energy mix, industrial structure, institutional quality, regulation, and environmental policy may still matter.
Comment 9
Reviewer comment: *Using both production-based and consumption-based CO2 emissions is a strength of the paper. However, the manuscript should more clearly explain what the differences between these two dependent variables imply for the pollution-haven, pollution-halo, and carbon-leakage arguments.*
Response: We thank the reviewer for recognizing this strength. The revised manuscript now explains that production-based emissions assign carbon to the territory where production occurs, while consumption-based emissions reallocate embodied carbon to final consumption. This distinction is important because territorial improvements may coexist

with imported or consumption-embodied emissions.
Manuscript change made: A new discussion was added explaining how the different results for production-based and consumption-based CO2 emissions affect the interpretation of pollution-haven, pollution-halo, and carbon-leakage mechanisms. The revised text avoids a simple pollution-halo conclusion and explains why FDI and trade effects depend on where emissions are accounted for.
Comment 10
Reviewer comment: *The authors should discuss whether the estimated average effects may conceal substantial country-level differences.*
Response: We agree. The revised manuscript now states that the reported coefficients are panel-average effects and may conceal substantial country-level heterogeneity in energy mix, institutional quality, sectoral structure, education systems, and environmental regulation.
Manuscript change made: A paragraph was added to the empirical discussion explaining the limits of panel-average interpretation. The limitations section was also expanded to recommend future testing of country-level, income-group, sectoral, and embodied-carbon heterogeneity.

Competing Interests: no

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